

Zaud Rehman

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SUMMARY

Backend software engineer with published open-source Rust crates and a portfolio spanning concurrent WebSocket backends, REST APIs, and lock-free data structures. Comfortable across the full stack, from Rust and Axum backend services to React and Three.js frontends, with a focus on performance and clean API design.

EDUCATION

Srinivas University

B.Tech in Computer Science Engineering | CGPA: 9.08/10.0

Mangaluru, Karnataka, India

Dec 2021 – Apr 2025

Karnataka Secondary Education Examination Board (KSEEB)

12th Grade, Percentage: 80%

Karnataka, India

2020

TECHNICAL SKILLS

Languages: Rust, Python, Java, C, C++, JavaScript/TypeScript, SQL

Frameworks & Technologies: Axum, FastAPI, Django REST, Next.js, WebSocket, REST APIs, React

Databases & Caching: PostgreSQL, MongoDB, Redis, SQLx

Cloud & Tools: Docker, Git, Oracle Cloud (OCI), JWT Authentication, Microservices, Cargo

Problem Solving: 675 problems solved on LeetCode; HackerRank Problem Solving (Intermediate) certified

PROJECTS

BloomCraft | Rust | GitHub | crates.io

Nov 2025 – Present

- Engineered a **lock-free concurrent Bloom filter library** supporting **12 variants** (Standard, Scalable, Partitioned, Sharded, etc.), achieving up to **80M insert ops/sec single-threaded** and **3.1× thread scaling to 62M at 8 threads** with measured FPR of **1% (Standard)** and as low as **0.02% (Scalable)** by implementing AtomicU64 CAS and RwLock-striped sharding with Lemire's fast range reduction.
- Reduced hash function calls from k to 2 per lookup via **Kirsch-Mitzenmacher Enhanced Double Hashing**, exposed through a **pluggable hash-strategy API** (FNV-1a default; WyHash, XXH3, SIMD via feature gates) and **type-state builder**, with optional HyperLogLog++ cardinality estimation and feature-gated serde support.
- Delivered a **production-ready crate** covering 12 Bloom filter variants via a single ergonomic API, targeting high-throughput **concurrent backends** and **data-pipeline** use cases at scale.

erra | Rust | GitHub | crates.io

Jun 2026 – Present

- Shipped a **zero-dependency, no_std-compatible** error-annotation library to crates.io, achieving **zero-allocation** on the static annotation path by using Cow's `static`, `str` and integrated `std::error::Error` chains.
- Maintained **MSRV 1.75.0** with **zero unsafe code**, enabling deployment to **embedded and WASM targets** without any std dependency, broadening library applicability across resource-constrained and browser runtimes.

Tributary | Python, Raft, TCP, WAL | GitHub

Feb 2026 – Jul 2026

- Engineered a **distributed, fault-tolerant message broker** from scratch using **Raft consensus**, segment-based **WAL**, and partitioned replication, with a custom binary wire protocol and TLS, on Python's standard library alone.
- Validated fault tolerance via **8 game-day failure drills** (leader failover, rolling restarts, WAL recovery) on a 3-node Docker cluster, achieving **335/335 passing CI/CD tests** under strict mypy typing.

NoteFlow | Rust, Axum, Next.js 14, PostgreSQL, Redis, WebSocket | Live | Backend

Aug – Dec 2025

- Scaled a **real-time collaborative backend** to **1,000+ concurrent WebSocket connections** at **sub-100ms latency**, by building a Rust/Axum service with **connection-pool tuning** and async task management.
- Reduced cross-device sync latency to **under 1 second** by implementing **Redis-based rate limiting**, JWT auth with **refresh-token rotation**, and a Markdown + LaTeX editor with **incremental state diffing**.

Aesthete | React 18, Three.js, Zustand, TailwindCSS | Live | GitHub

Jan 2026 – Present

- Built a **3D algorithm visualization engine** rendering **25+ algorithms** (DP, Graphs, Sorting) at a **steady 60 FPS**, by leveraging WebGL, Three.js **scene-graph batching**, and Zustand for **minimal re-render overhead**.
- Improved debuggability by designing a **generator-based AlgorithmController** supporting **time-travel debugging** and **frame-perfect stepping**, with synchronized multi-language code highlighting for **educational tooling** use cases.

INTERNSHIPS

Backend Development Intern

May 2024 – Jun 2024

Tikanga Pvt. Limited

Mangaluru, Karnataka, India

- Reduced API bug rate by **15%** across **user management, Auth, and business-logic endpoints** built on Django REST Framework, by introducing systematic unit and integration testing and participating in structured code reviews.

Artificial Intelligence & Machine Learning Intern

Sep 2023

Thaniya Technologies

Mangaluru, Karnataka, India

- Improved model prediction accuracy by **10%** across a **10,000+ record** dataset, by building scikit-learn preprocessing pipelines with feature engineering, normalization, and systematic **hyperparameter tuning** via cross-validation.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Data Science Professional – Oracle (2025)

Problem Solving (Intermediate) – HackerRank (2025)

AWS Artificial Intelligence Practitioner Learning – Amazon Web Services (2025)